

Complex Analysis

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Contents

1	Complex Analysis (複分析 or 複變分析)	1
I	Derivative	1
i	Derivative	1
ii	Smoothness.	1
iii	Cauchy–Riemann equations	1
iv	Holomorphic or analytic function	2
v	Biholomorphic function	2
II	Poles and Zeros	2
i	Poles	2
ii	Zeros	2
iii	Meromorphic function.	2
iv	Pole-zero constellation or plot	2
III	Essential singularity	3
IV	Annulus	3
V	Contour	3
VI	Contour integral.	3
i	Cauchy integral theorem or Cauchy–Goursat theorem.	3
ii	Cauchy’s integral formula	3
iii	Laurent series.	3
iv	(Cauchy’s) residue theorem	4
v	Jordan’s lemma	4
vi	Estimation lemma.	5

1 Complex Analysis (複分析 or 複變分析)

I Derivative

i Derivative

Let W be a topological vector space. The ordinary derivative or derivative of a function $f(u): U \subseteq \mathbb{C} \rightarrow W$ (with respect to u) at $x \in U$, denoted as $f'(x)$, is defined as

$$f'(x) = \lim_{z \rightarrow x} \frac{f(z) - f(x)}{z - x}$$

if the limit exists. If such limit exists, we say f is (complex) differentiable at x .

We define the (first(-order)) ordinary derivative (function) or derivative (function) of f (with respect to u) as a function f' with codomain W such that for any $x \in U$ at which f is (complex) differentiable, f' maps x to the derivative of f at x .

The derivative of the k th(-order) ($k \in \mathbb{N}$) derivative function of f at $x \in U$ is called the $(k+1)$ th(-order) derivative of f (with respect to u) at $x \in U$. The derivative function of the k th(-order) ($k \in \mathbb{N}$) derivative function of f is called the $k+1$ th(-order) derivative function of f (with respect to u).

If for any $n \in \mathbb{N}$, the n th(-order) derivative of a function f at a point x in its domain exists, we say f is infinitely (complex) differentiable at x .

If f is (complex) differentiable at all point in $I \subseteq U$, we say f is (complex) differentiable on I ; if f is (complex) differentiable on U , we say f is (complex) differentiable. If $f^{(n-1)}$ exists and is (complex) differentiable at all point in $I \subseteq U$, we say f is n -times (complex) differentiable on I ; if $f^{(n-1)}$ exists and is (complex) differentiable on U , we say f is n -times (complex) differentiable.

The operation of finding the derivative or derivative function is called ordinary differentiation or differentiation.

Specifically, the 0th(-order) derivative of f is f itself.

ii Smoothness

Same as that of real.

iii Cauchy–Riemann equations

A function $f(x + iy) = u(x, y) + iv(x, y)$ for some real-valued functions u, v with $D_u, D_v \subseteq \mathbb{R}_2$ is holomorphic if u and v satisfy the Cauchy–Riemann equations:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x}.\end{aligned}$$

The converse is true if f is continuous.

Proof. TODO

□

iv Holomorphic or analytic function

A function f is holomorphic or analytic on $U \subseteq D_f$ if it is complex differentiable at every $z \in U$.

A function f is holomorphic or analytic if it is holomorphic on D_f .

A function f is holomorphic or analytic at $z \in D_f$ if it is holomorphic on some open neighborhood of z .

v Biholomorphic function

A function f is biholomorphic on $U \subseteq D_f$ iff $f|_U$ is bijective and holomorphic and $(f|_U)^{-1}$ is holomorphic.

A function is biholomorphic iff it is bijective and holomorphic and its inverse is holomorphic.

II Poles and Zeros

i Poles

An isolated singularity z_0 of a complex function $f(z)$ is a pole of it iff there exists $n \in \mathbb{N}$, called the order, multiplicity, or degree of z_0 , such that

$$(z - z_0)^n f(z)$$

is holomorphic on an open neighborhood of z_0 and

$$\lim_{z \rightarrow z_0} (z - z_0)^n f(z) \neq 0.$$

A simple or single pole is a pole of order 1. A repeated or multiple pole is a pole of order > 1 .

ii Zeros

A point is a zero of a complex function f iff it is a pole of $\frac{1}{f}$.

A point is a zero of order, multiplicity, or degree $-n$ of a complex function f iff it is a pole of order n of $\frac{1}{f}$.

A simple or single zero is a zero of order -1. A repeated or multiple zero is a zero of order < -1 .

iii Meromorphic function

A complex function f is meromorphic on an open set U if for all $z \in U$, either f is holomorphic at z or z is a pole of f .

A complex function f is meromorphic if it is meromorphic on D_f .

iv Pole-zero constellation or plot

A scattering plot on the complex plane showing the poles and zeros of a meromorphic function.

III Essential singularity

An isolated singularity of a complex function is an essential singularity iff it is neither a removable singularity nor a pole.

IV Annulus

An annulus $\text{ann}(a; r, R)$ with $R \leq \infty$ in the complex plane is a region defined as

$$r < |z - a| < R.$$

If $r = 0$, the region is also called punctured disk.

V Contour

A contour is a parameterized curve $\gamma(t) : [a, b] \rightarrow \mathbb{C}$ that is piecewise C^1 .

VI Contour integral

The contour integral of a complex function f defined on the image of a contour $\gamma(t) : [a, b] \rightarrow \mathbb{C}$ over γ is defined as the Riemann–Stieltjes integral:

$$\int_{\gamma} f(z) dz := \int_{t=a}^b f(\gamma(t)) d\gamma(t).$$

i Cauchy integral theorem or Cauchy–Goursat theorem

TODO

ii Cauchy's integral formula

TODO

iii Laurent series

Let f be holomorphic on an annulus $A = \text{ann}(z_0; r; R)$. The Laurent series of f at z_0 is

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n + \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}, \quad z \in A,$$

where the first term is called regular or analytic part and the second term is called principal or singular part, and a_n is given by

$$a_n = \frac{1}{2\pi i} \oint_{\gamma} \frac{f(z)}{(z - z_0)^{n+1}} dz$$

for any closed contour $\gamma \in A$ enclosing z_0 .

- $\forall n \in \mathbb{N} : a_{-n} = 0$: either f is holomorphic at z_0 or z_0 is a removable singularity of f .
- $\exists N \in \mathbb{N}$ s.t. $a_{-N} \neq 0 \wedge \forall n \in \mathbb{N}_{>N} : a_{-n} = 0$: z_0 is a pole of order N of f .

- Otherwise: z_0 is a essential singularity of f .

a_{-1} is called residue of f at z_0 , denoted as $\text{Res}(f, z_0)$ or $\text{Res}(f)$, that is,

$$\text{Res}(f, z_0) = \frac{1}{2\pi i} \oint_{\gamma} f(z) dz.$$

iv (Cauchy's) residue theorem

Let U be a simply connected open subset of the complex plane containing a finite list of points $a_1, \dots, a_n$, $U_0 = U \setminus \{a_1, \dots, a_n\}$, and a function f is holomorphic on U_0 . Let γ be a closed rectifiable curve in U_0 . Let the winding number of γ around a_k be $I(\gamma, a_k)$.

$$\oint_{\gamma} f(z) dz = 2\pi i \sum_{k=1}^n I(\gamma, a_k) \text{Res}(f, a_k).$$

Proof. By divergence theorem, obviously. □

v Jordan's lemma

Consider a semicircular contour

$$C_R = \{Re^{i\theta} : \theta \in [0, \pi]\}$$

If $f(z)$ is a function of the form

$$f(z) = e^{iaz} g(z), \quad z \in \mathbb{C}$$

with a positive parameter $a > 0$, then Jordan's lemma states that

$$\left| \int_{C_R} f(z) dz \right| \leq \frac{\pi}{a} M_R,$$

where

$$M_R := \max_{\theta \in [0, \pi]} |g(Re^{i\theta})|.$$

Proof.

$$\begin{aligned}
\left| \int_{C_R} f(z) dz \right| &= \left| \int_0^\pi g(Re^{i\theta}) e^{iaR(\cos \theta + i \sin \theta)} i Re^{i\theta} d\theta \right| \\
&= R \left| \int_0^\pi g(Re^{i\theta}) e^{aR(i \cos \theta - \sin \theta)} i e^{i\theta} d\theta \right| \\
&\leq R \int_0^\pi \left| g(Re^{i\theta}) e^{aR(i \cos \theta - \sin \theta)} i e^{i\theta} \right| d\theta \\
&= R \int_0^\pi |g(Re^{i\theta})| e^{-aR \sin \theta} d\theta \\
&\leq RM_R \int_0^\pi e^{-aR \sin \theta} d\theta \\
&= 2RM_R \int_0^{\frac{\pi}{2}} e^{-aR \sin \theta} d\theta, \quad \sin \theta \geq \frac{2\theta}{\pi} \\
&\leq 2RM_R \int_0^{\frac{\pi}{2}} e^{-\frac{2aR\theta}{\pi}} d\theta \\
&= 2RM_R \left\{ -\frac{\pi}{2aR} e^{-\frac{2aR\theta}{\pi}} \right\} \Big|_0^{\frac{\pi}{2}} \\
&= \frac{\pi}{a} M_R (1 - e^{-aR}) \\
&\leq \frac{\pi}{a} M_R
\end{aligned}$$

□

If f is integrable on C_R for all sufficiently large R and

$$\lim_{R \rightarrow \infty} M_R = 0$$

then by Jordan's lemma

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0$$

vi Estimation lemma

TODO